

JONATHAN KIM

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Education

The University of Texas at El Paso

Expected: May 2027

Bachelor of Science in Computer Science

GPA: 3.57/4.00

Relevant Coursework: Machine Learning, Software Engineering, Data Structures & Algorithms, Database Systems, Operating Systems, Computer Organization, Computer Security

Activities: Secretary, Free and Open-Source Club (2024–2025)

Experience

Software Engineer (Part-time)

November 2025 – Present

PM Technologies LLC

El Paso, Texas

- Replaced hourly manual floor checks with a Flask API monitoring 7 networked 3D printers, covered by 203 tests
- Eliminated silent record duplication traced to FAT32 filename truncation on printer storage with a targeted dedup fix
- Prevented SQLite data corruption from power loss with write-ahead-log backups and atomic snapshot swaps
- Enabled safe, repeatable schema migrations on a live Raspberry Pi with idempotency guards and pre-migration backups

Undergraduate Teaching Assistant

January 2025 – May 2025

The University of Texas at El Paso

El Paso, Texas

- Mentored 30+ students through Java labs, debugging, and object-oriented design

Machine Learning Researcher

May 2024 – July 2024

The Ohio State University AI-EDGE Institute

Columbus, Ohio

- Cut Stable Diffusion 2 fine-tuning 55% (11h to 5h) with LoRA adapters

Projects

Sophia AI

January 2026 – April 2026

- Built an SMS assistant turning natural-language texts into calendar events, covered by 115 tests
 - Eliminated duplicate sends under concurrent workers and retries with SKIP LOCKED and SID keys
- Python • FastAPI • PostgreSQL • Twilio • Stripe • Railway • Claude API

BorderETA | bordereta.live | 800+ users

December 2025 – Present

- Launched a bilingual iOS/Android app for commuters at 5 international border crossings
 - Eliminated crossing data loss in cellular dead zones with a write-ahead queue replayed on relaunch
 - Swapped battery-draining background polling for 10 nested geofences under iOS's 20-region cap
 - Reduced wait-time prediction error 40% with a quantile forecaster on FastAPI
- React Native • Expo • TypeScript • Supabase

ReconAIssance | co-founded | 2nd Place, AI Hackathon

October 2025 – March 2026

- Detected weapons and persons of interest by training ArcFace and fine-tuning YOLOv8 with hard-negative mining
 - Delivered per-frame risk scoring under 200ms by fusing model outputs in one async pass
- Python • FastAPI • OpenCV • YOLOv8 • InsightFace • React

DUI Risk Radar | github.com/jonkim13/DUI-Risk-Radar | 2nd Place, BorderHack

September 2024

- Led a team of 3 to ship a real-time map of DUI incidents and high-risk zones
- JavaScript • React • Firebase • HTML/CSS

Technical Skills

Languages: Python, Java, TypeScript, JavaScript, SQL

Frameworks: FastAPI, Flask, React, React Native, PyTorch

Data & Cloud: PostgreSQL, SQLite, Supabase (SQL), Firebase (NoSQL), Docker, Git/GitHub, CI/CD, Linux, pytest